Math 241, F18
Exam 2

Name: Answer

<u>**Instructions:**</u>

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the exam.

Problem	Points	Score
1	20	
2	20	
3	25	
4	20	
5	20	
Total	105	

Good Luck !

Problem 1. (20 points)

(a) Use Chain Rule to find $\dfrac{\partial z}{\partial s}$ and $\dfrac{\partial z}{\partial t}$ in terms of s and t when $z = \ln(2x + 3y)$, $x = te^s$, $y = e^{st}$.

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s} = \frac{2}{2x+3y}(te^s) + \frac{3}{2x+3y}(te^{st})$$

$$= \frac{2te^s}{2te^s+3e^{st}} + \frac{3te^{st}}{2te^s+3e^{st}} = \frac{2te^s+3te^{st}}{2te^s+3e^{st}}$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t} = \frac{2}{2x+3y}(e^s) + \frac{3}{2x+3y}(se^{st})$$

$$= \frac{2e^s}{2te^s+3e^{st}} + \frac{3se^{st}}{2te^s+3e^{st}} = \frac{2e^s+3se^{st}}{2te^s+3e^{st}}$$

(b) Given $f(x, y) = x^2 - 4xy + 2y^2$, $\mathbf{p} = (-1, 2)$ and $\mathbf{a} = \mathbf{i} - \sqrt{3}\mathbf{j}$, find the directional derivative of f at the point $\mathbf{p}$ in the direction of $\mathbf{a}$.

$$f_x = 2x - 4y, \quad f_y = -4x + 4y$$

$$\Rightarrow \nabla f(-1, 2) = \langle 2\cdot(-1) - 4\cdot 2, (-4)\cdot(-1) + 4\cdot 2 \rangle$$

$$= \langle -10, 12 \rangle$$

$$a = \langle 1, -\sqrt{3} \rangle \Rightarrow u = \frac{a}{|a|} = \frac{\langle 1, -\sqrt{3}\rangle}{2} = \langle \tfrac{1}{2}, -\tfrac{\sqrt{3}}{2} \rangle$$

$$\Rightarrow D_u f(-1, 2) = \langle -10, 12 \rangle \cdot \langle \tfrac{1}{2}, -\tfrac{\sqrt{3}}{2} \rangle$$

$$= -5 - 6\sqrt{3}$$

Problem 2. (20 points)
(a) Assume that the equation $xy+y^2-3x-3=0$ defines y as a function of x. Using **implicit differentiation** to find $\frac{dy}{dx}$ at the point $(-1,1)$. (*Hint:* $\frac{dy}{dx} = -\frac{F_x}{F_y}$)

Let $F(x,y)=xy+y^2-3x-3$

$F_x = y-3, \quad F_y = x+2y$

$\Rightarrow F_x(-1,1) = 1-3 = -2$

$F_y(-1,1) = -1+2\cdot 1 = 1$

Thus $\left.\frac{dy}{dx}\right|_{(1,2)} = -\frac{F_x(-1,1)}{F_y(-1,1)} = 2$

(b) If $f(x,y) = ye^{xy}$, find an approximation of $f(1.1,-0.05)$ by using the **linear approximation** for f at (1,0).

$f_x = y^2e^{xy}, \quad f_y = e^{xy}+xye^{xy}$

$\Rightarrow f_x(1,0)=0, \quad f_y(1,0)=1$

$f(x,y) \approx L(x,y) = f(1,0) + f_x(1,0)(x-1) + f_y(1,0)(y-0)$

$= 0+0\cdot(x-1)+1\cdot y$

$= y$

Thus $f(1.1,-0.05) \approx -0.05$

Problem 3. (25 points)
(a) Find all critical points of $f(x, y) = x^2y - 6y^2 - 3x^2$ and then determine whether each such a point gives a local maximum or minimum, or whether it is a saddle point.

$$\left\{\begin{array}{l} f_x = 2xy - 6x = 0 \Rightarrow 2x(y-3) = 0 \\ f_y = x^2 - 12y = 0 \Rightarrow y = \frac{x^2}{12} \end{array}\right\} \Rightarrow \begin{array}{l} 2x(\frac{x^2}{12} - 3) = 0 \\ x(x^2-36) = 0 \\ x(x+6)(x-6) = 0 \\ x = 0,\ x = -6,\ x = 6. \\ y = 0 \quad y = 3 \quad y = 3 \end{array}$$

critical points: $(0,0), (-6,3), (6,3)$

$f_{xx} = 2y - 6$, $f_{yy} = -12$, $f_{xy} = 2x$

$D = f_{xx}f_{yy} - f_{xy}^2$

For $(0,0)$, $f_{xx}(0,0) = -6 < 0$, $D(0,0) = (-6)(-12) - 0 = 72 > 0$

$\Rightarrow (0,0)$ gives a local maximum value 0.

For $(6,-3)$, $D(-6,3) = (0)\cdot(-12) - (-12)^2 = -144 < 0$

$\Rightarrow (-6,3)$ is a saddle point

For $(6,3)$, $D(6,3) = 0\cdot(12) - (12)^2 = -144 < 0$

$\Rightarrow (6,3)$ is a saddle point

(b) Use the method of **Lagrange multipliers** to find the maximum and minimum value of the function $f(x, y) = x + 3y$ subject to the constraint $x^2 + y^2 = 40$.

$\nabla f = \langle 1, 3 \rangle$, $\nabla g = \langle 2x, 2y \rangle$ where $g(x,y) = x^2 + y^2$

$$\left\{\begin{array}{l} \nabla f = \lambda \nabla g \\ g(x,y) = 40 \end{array}\right. \Rightarrow \left\{\begin{array}{l} 1 = \lambda\cdot(2x) \Rightarrow x = \frac{1}{2\lambda} \\ 3 = \lambda\cdot(2y) \Rightarrow y = \frac{3}{2\lambda} \\ x^2 + y^2 = 40 \end{array}\right\} \Rightarrow \begin{array}{l} (\frac{1}{2\lambda})^2 + (\frac{3}{2\lambda})^2 = 40 \\ \frac{10}{4\lambda^2} = 40 \\ \lambda^2 = \frac{1}{16} \Rightarrow \lambda = \pm\frac{1}{4} \end{array}$$

$\lambda = \frac{1}{4} \Rightarrow x = 2,\ y = 6 \Rightarrow f(2,6) = 2 + 3\cdot 6 = 20$

$\lambda = -\frac{1}{4} \Rightarrow x = -2,\ y = -6 \Rightarrow f(-2,-6) = (-2) + 3\cdot(-6) = -20$

$\Rightarrow$ The maximum value is 20,

The minimum value is -20.

Problem 4. (20 points) Evaluate the following iterated integrals.

(a) $\int_0^1 \int_1^2 xye^y \, dxdy$

$= \int_0^1 \left[\frac{x^2}{2} y e^y\right]_{x=1}^{x=2} dy$

$= \int_0^1 \frac{3}{2} y e^y \, dy$

$= \frac{3}{2} \int_0^1 y e^y \, dy$

$= \frac{3}{2} \left[y e^y \Big|_0^1 - \int_0^1 e^y \, dy \right]$

$= \frac{3}{2} \left[e - \left[e^y\right]_{y=0}^{y=1} \right]$

$= \frac{3}{2} \left[e - (e-1) \right]$

$= \frac{3}{2}$

(b) $\int_0^1 \int_0^x \frac{2y}{x^3+1} \, dydx$

$= \int_0^1 \left[\frac{y^2}{x^3+1} \right]_{y=0}^{y=x} dx$

$= \int_0^1 \frac{x^2}{x^3+1} \, dx$

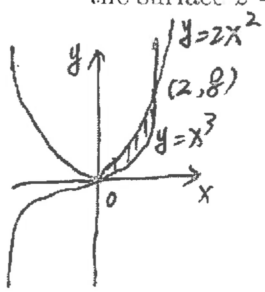
$= \left[\frac{1}{3} \ln(x^3+1) \right]_{x=0}^{x=1}$

$= \frac{1}{3} \ln 2 - \frac{1}{3} \ln 1$

$= \frac{1}{3} \ln 2$

Problem 5. (20 points)

(a) Let R be the region bounded by $y = 2x^2$ and $y = x^3$. Find the volume of the solid under the surface $z = 2x + y$ and above the region R by evaluating $\iint_R 2x + y\, dA$.



$$\begin{cases} y=2x^2 \\ y=x^3 \end{cases} \Rightarrow x^3-2x^2=0 \Rightarrow x^2(x-2)=0 \Rightarrow \begin{matrix} x=0, & x=2 \\ \Downarrow & \Downarrow \\ y=0 & y=8 \end{matrix}$$

Then $R = \{(x,y) \mid 0 \le x \le 2,\ x^3 \le x \le 2x^2\}$ ← TYPE I

$$\iint_R 2x+y\,dA = \int_0^2 \int_{x^3}^{2x^2} 2x+y\,dy\,dx$$

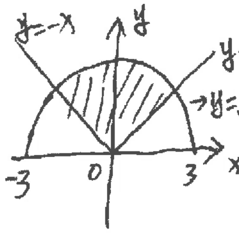
$$= \int_0^2 \left[2xy + \frac{y^2}{2}\right]_{y=x^3}^{y=2x^2} dx$$

$$= \int_0^2 \left[4x^3 + \cancel{2x^4} - \cancel{2x^4} - \frac{x^6}{2}\right] dx$$

$$= \left[x^4 - \frac{1}{14}x^7\right]_{x=0}^{x=2}$$

$$= 16 - \frac{128}{14} = \frac{48}{7}$$

(b) A lamina covering the region R in the plane is bounded by the curves $y = x$, $y = -x$ and $y = \sqrt{9-x^2}$. Assume it has a density $f(x,y) = \sqrt{x^2+y^2}$. Find the mass of the lamina by evaluating the double integral $\iint_R \sqrt{x^2+y^2}\, dA$ in the **polar coordinates**.



$R = \{(r,\theta) \mid 0 \le r \le 3,\ \frac{\pi}{4} \le \theta \le \frac{3}{4}\pi\}$

$$\iint_R \sqrt{x^2+y^2}\,dA = \int_{\frac{\pi}{4}}^{\frac{3}{4}\pi} \int_0^3 r \cdot \underline{r\,dr\,d\theta}$$

$$= \int_{\frac{\pi}{4}}^{\frac{3}{4}\pi} \left[\frac{r^3}{3}\right]_{r=0}^{r=3} d\theta$$

$$= \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} 9\,d\theta$$

$$= 9 \cdot \left(\frac{3}{4}\pi - \frac{\pi}{4}\right)$$

$$= \frac{9}{2}\pi$$